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LIST OF ABBREVIATION

Number	Abbreviation	Meaning
1	WGI	World Governance Indicators
2	GMM	generalized method of moments
3	SDGs	Sustainable Development Goals
4	IEI	Innovation Efficiency Index
5	GEE	Governance effectiveness
6	VAE	Voice and Accountability
7	PVE	Political Stability and Absence of Violence
8	RQE	Regulatory Quality
9	RLE	Rule of Law
10	CCE	Control of Corruption
11	PPP	GDP-per-capita
12	NRR/GDP	Natural Resource Rents as a Percentage of GDP
13	NFDI	Net Foreign Direct Investment
14	Pop	Population
15	HS	Health Spending
16	LE	Life expectancy at birth
17	DVLP	Developing country status
18	OLS	Ordinary Least Squares
19	SSR	Standard Regression Results

Part1: Overview and Data Description

1. Overview of Topic:

Governance quality and institutional effectiveness are critical factors influencing economic growth. This section synthesizes insights from three recent studies (2023–2024) that explore the relationship between governance measures and economic performance, identifying which governance impacts growth of a country.

a. Research Articles:

1) Lopes, Packham, and Walther (2023):

This study examines the influence of six World Governance Indicators (WGIs) on the GDP growth of BRICS countries as well as some developed economies. Some key variables considered in their analysis include regulatory quality, rule of law, governance effectiveness, and control of corruption. The findings reveal that these governance indicators have a positive impact on economic growth, highlighting their role in fostering efficient markets and business environments.

2) OECD (2023):

The OECD (2023) report discusses structural reform priorities to foster sustainable long-term growth across OECD and partner countries. The key variables considered in this analysis include the employment rate, labor efficiency, gender inequality, and working age. The findings highlight that structural reforms aimed at addressing inequality and sustainability are instrumental in driving economic growth. Governance quality is emphasized as a critical enabler, underscoring the importance of effective governance in achieving these reforms and promoting sustainable development.

3) Hussen (2023):

This study analyzes institutional quality in 31 Sub-Saharan African countries using a two-step system-GMM technique. The key variables examined in this analysis include investment-promoting institutions, democratic institutions, and conflict-preventing institutions. The findings indicate that both investment-promoting and democratic institutions have a positive influence on economic growth.

b. Paper Selection:

Lopes, Packham, and Walther (2023) provide a comprehensive examination of governance impacts across emerging and developed economies. Their analysis includes variables such as regulatory quality, rule of law, debt, budget balance, current account balance, trade balance, foreign exchange

rate, and interest rates. The study highlights that these variables are significant for macroeconomic stability and growth. The findings indicate that while governance factors vary across countries, they remain stable over time, suggesting opportunities for targeted policy interventions.

c. Variables for Analysis:

Drawing from the reviewed literature, the following variables are selected for further analysis:

- 1) Governance indicators: Governance effectiveness, and control of corruption.
- 2) Macroeconomic variables: GDP-per-capita, and net foreign direct investment.
- 3) Social indicators: Health spending, and natural resource rents.

d. Link to SDGs:

Governance and institutional quality promote transparent, accountable, and inclusive institutions, which support various SDGs by fostering economic stability, reducing inequality, and enabling innovation. For example, regulatory quality enhances business innovation and economic inclusivity, aligning with SDG 8 (Decent Work and Economic Growth). Additionally, strengthening institutional frameworks contributes to SDG 9 (Industry, Innovation, and Infrastructure) by fostering healthy growth for countries.

e. Research Questions:

- 1) How does government effectiveness influence GDP-per-capita in developing countries?

This question aligns with the literature's emphasis on governance as a catalyst for sustainable growth in resource-constrained environments, particularly for achieving SDG 9.

- 2) How do governance measures, such as health spending (% of GDP), natural resource rents, and control of corruption, enhance innovation efficiency and contribute to sustainable economic development?

This question explores how these variables influence GDP-per-capita, emphasizing governance's role in fostering innovation and inclusivity, and aligning with SDG 8.

2. Data Importation:

a. Descriptive Statistics:

Variable	Mean	Median	SD	Min	Max	N
IEI	33.79	31.25	14.43	0.00	73.60	224
GEE	-0.08	-0.31	0.90	-1.29	2.24	224
VAE	-0.20	-0.20	0.69	-1.66	1.62	224
PVE	-0.40	-0.50	0.99	-2.60	1.62	224
RQE	-0.11	-0.27	0.85	-1.90	2.26	224
RLE	-0.15	-0.34	0.82	-1.58	2.01	224
CCE	-0.21	-0.44	0.87	-1.40	2.34	224
GDP-per-capita (\$)	14,306.43	5,122.25	20,164.33	1,089.87	98,455.33	224
NRR/GDP (%)	5.99	4.52	5.51	0.00	22.51	224
NFDI (\$)	14,140,401,382	1,897,522,693	41,221,234,060	-4,156,404,703	290,928,431,467	224
Pop	136,210,334	25,718,552	321,193,313	404,421	1,397,715,000	224
Inflation (%)	4.22	2.64	5.89	-17.61	52.99	224
HS (%)	2.21	1.74	1.85	0.15	9.21	224
LE (Years)	67.90	66.66	8.28	52.23	84.21	224
DVLP	0.84	1.00	0.36	0.00	1.00	224

Table1: Descriptives Statistics of 15 variables

The descriptive statistics table provides a summary of the central tendency, dispersion, and range of the variables in the dataset. For example:

- GDP-per-capita (PPP) has a mean of 14,306.43 with high variability (SD = 20,164.33), reflecting significant income disparities across countries.

- Innovation Efficiency Index (IEI) has a mean of 33.79 and a wide range from 0 to 73.6, indicating differences in innovation outputs relative to inputs.
- Developing has a mean of 0.84, suggesting that 84% of the countries in the dataset are developing.

b. Correlation Table:

Variable	PPP	IEI	NFD I	Pop	NRR /GD P	Infla tion	HS	LE	DVL P	GEE	VAE	PVE	RQE	RLE	CCE
PPP	1.00	0.55	0.25	-0.09	-0.16	-0.29	0.44	0.74	-0.88	0.87	0.27	0.69	0.86	0.82	0.83
IEI	0.55	1.00	0.34	0.14	-0.32	-0.07	0.52	0.58	-0.48	0.67	0.37	0.49	0.67	0.65	0.63
NFDI	0.25	0.34	1.00	0.73	-0.25	-0.11	0.09	0.34	-0.10	0.29	-0.25	0.11	0.18	0.15	0.18
Pop	-0.09	0.14	0.73	1.00	-0.25	-0.05	-0.06	0.15	0.13	0.06	-0.16	-0.16	-0.08	-0.04	-0.07
NRR/G DP	-0.16	-0.32	-0.25	-0.25	1.00	0.16	-0.29	-0.48	0.13	-0.36	-0.26	0.00	-0.31	-0.35	-0.28
Inflation	-0.29	-0.07	-0.11	-0.05	0.16	1.00	-0.13	-0.05	0.16	0.11	-0.02	0.03	0.07	0.08	0.09
HS	0.44	0.52	0.09	-0.06	-0.29	-0.13	1.00	0.39	-0.41	0.46	0.14	0.33	0.48	0.47	0.43
LE	0.74	0.58	0.34	0.15	-0.48	-0.05	0.39	1.00	-0.78	0.78	0.33	0.59	0.82	0.83	0.80
DVLP	-0.88	-0.48	-0.10	0.13	0.13	0.16	-0.41	-0.78	1.00	-0.83	-0.31	-0.61	-0.85	-0.84	-0.83
GEE	0.87	0.67	0.29	0.06	-0.36	0.11	0.46	0.78	-0.83	1.00	0.32	0.67	0.89	0.88	0.85
VAE	0.27	0.37	-0.25	-0.16	-0.26	-0.02	0.14	0.33	-0.31	0.32	1.00	0.29	0.34	0.35	0.33
PVE	0.69	0.49	0.11	-0.16	0.00	0.03	0.33	0.59	-0.61	0.67	0.29	1.00	0.72	0.74	0.70
RQE	0.86	0.67	0.18	-0.08	-0.31	0.07	0.48	0.82	-0.85	0.89	0.34	0.72	1.00	0.92	0.88
RLE	0.82	0.65	0.15	-0.04	-0.35	0.08	0.47	0.83	-0.84	0.88	0.35	0.74	0.92	1.00	0.91
CCE	0.83	0.63	0.18	-0.07	-0.28	0.09	0.43	0.80	-0.83	0.85	0.33	0.70	0.88	0.91	1.00

Table2: Correlation between 15 variables

The correlation table reveals relationships among variables:

- PPP positively correlates with Life Expectancy (0.74) and Governance Effectiveness (0.87), suggesting that wealthier countries tend to have better health outcomes and governance.
- IEI shows a moderate positive correlation with Health spending (0.52, indicating that innovation is linked to institutional quality and resource allocation.
- Total-Natural-Rent/GDP is negatively correlated with most governance indicators, highlighting the potential "resource curse."

c. Two Governance Indicators:

1) Control of Corruption (CCE):

CCE reflects perceptions of the extent to which public power is exercised for private gain, including petty and grand corruption, as well as state capture. Corruption undermines economic efficiency, distorts the allocation of resources, and reduces investor confidence, all of which can hinder economic growth. The CCE metric is sourced from the World Bank's WGI and is derived from survey data collected from citizens, businesses, and experts.

2) Government Effectiveness (GEE):

GEE measures the quality of public services, the capacity of the civil service, and the government's ability to formulate and implement sound policies. Efficient governance fosters economic stability, encourages innovation, and enhances long-term growth. The GEE metric is sourced from the World Bank's WGI and is compiled from various data sources, including surveys of businesses and households.

Part2: Initial Estimation:

1. Expectations for Signs of Coefficients:

1) Control of Corruption:

CCE is expected to have a positive impact on economic growth. Supporting literature, such as the study by Lopes, Packham, and Walther (2023), found that corruption control positively influences macroeconomic stability and GDP growth by improving governance.

2) Total-Natural-Rent/GDP:

TNR/GDP is expected to have a negative impact on economic growth. Economies that rely heavily on resource rents often experience slower long-term growth due to the "resource curse" phenomenon. Poor resource management and weaker governance structures in resource-dependent countries exacerbate this issue, leading to inefficiencies and reduced economic progress (Lopes, Packham, and Walther, 2023).

3) Net Foreign Direct Investment (NFDI):

NFDI is expected to have a positive impact on economic growth. Higher NFDI inflows typically drive capital investment, technology transfer, and job creation, all of which stimulate economic growth. This is supported by findings in OECD (2023).

Dependent Variable	Gdp-per-captita (Model 1)	Gdp-per-captita (Model 2)
Intercept	18325.1 (***)	15270.0 (***)
CCE	19170.1 (***)	19330.0 (***)
Total Natural Rent (% GDP)	-	373.8 (**)
NFDI	-	6.046e-08 (**)
R ²	0.683	0.702
Adjusted R ²	0.681	0.698
Residual Std. Error	11,380	11,070

Table3: Comparison between Models 1 and 2 using OLS.

Model 1 provides a focused understanding of how governance effectiveness alone influences GDP-per-capita, yielding an R² of 0.683, indicating good explanatory power. Model 2 offers a more comprehensive view by incorporating macroeconomic factors such as resource dependency and foreign investment, enabling a broader analysis of the determinants of economic growth, with the results estimated showing a higher R² of 0.702.

2. Proposed Adjustments:

- Log-transformation of GDP-per-capita: Since PPP follows a non-linear distribution, applying a log-transformation could normalize the data and improve interpretability (appendix 1,2).
- Omitting variable: The NFDI coefficient was found to be extremely small, indicating a negligible practical contribution to the model. This adjustment is expected to simplify the model while maintaining its explanatory power, thereby providing more robust and actionable insights into the relationship between the remaining variables and the dependent variable.

Dependent Variable	Gdp-per-captita (Model 1)	Gdp-per-captita (Model 2)	log (Gdp-per-captita) (Model 3)
Intercept	18,325.080 (***) (781.949)	15,265.150 (***) (1,191.264)	9.214 (***) (0.073)
CCE	19,170.150 (***) (876.202)	19,329.460 (***) (894.860)	1.019 (***) (0.059)
Total Natural Rent (% GDP)	-	373.836 (***) (143.530)	-0.030 (***) (0.009)
NFDI	-	0.00 (***) (0.00)	-
Observations	224	224	224
R²	0.683	0.702	0.625
Adjusted R²	0.682	0.698	0.622
Residual Std. Error	11,375.680 (df = 222)	11,073.300 (df = 220)	0.735 (df = 221)
F Statistic	478.677 (df = 1, ***)	173.155 (df = 3, ***)	184.499 (df = 2, ***)

Table4: Results comparison for three models

Part3: Interpretation

1. Regression Results in Estimated Equation Format:

Model 1: $\text{GDP-per-capita} = 18,325.08 + 19,170.15 \times \text{CCE}$

Model 2: $\text{GDP-per-capita} = 15,265.15 + 19,329.46 \times \text{CCE} + 373.83 \times \text{Total-Natural-Rent} + 0.00 \times \text{NFDI}$

Model 3: $\log(\text{GDP-per-capita}) = 9.214 + 1.019 \times \text{CCE} - 0.03 \times \text{Total-Natural-Rent}$

2. Best-Performing Model:

Model 3 is the best-performing model because:

- 1) The log-transformation stabilizes variance, making coefficients interpretable as elasticities (percentage changes).
- 2) All explanatory variables in Model 3 are statistically significant at the 1% level, which strengthens confidence in the results.
- 3) Though the Adjusted R² (0.622) is slightly lower than Model 2's R², Model 3 demonstrates better residual behavior (RSE = 0.735) and aligns more closely with the theoretical framework of the research question.

3. Evaluation of Preferred Model:

a. Statistical Significance (t-tests):

Hypotheses for Each Coefficient:

- Null Hypothesis (H_0): Coefficient = 0 (no relationship exists).
- Alternative Hypothesis (H_A): Coefficient $\neq 0$ (a relationship exists).

Variable	Coefficient	Std. Error	t-value	p-value	Significance Level	Result
Intercept	9.214	0.073	126.22	<0.01	1%	Reject H_0
CCE	1.019	0.059	17.27	<0.01	1%	Reject H_0
Total-Natural-Rent	-0.030	0.009	-3.33	<0.01	1%	Reject H_0

Table5: Hypothesis testing results

b. Interpretation of Coefficients:

1) Intercept:

When all predictors are zero, the log-transformed GDP-per-capita is approximately 9.214. This serves as the baseline value.

2) CCE:

A 1-unit increase in CCE is associated with a **1.019% increase in GDP-per-capita**, holding other factors constant. This reflects a strong, positive elasticity, indicating that higher CCE significantly boosts economic outcomes.

3) Total -Natural-Rent/GDP:

A 1% increase in total-natural-rent as a share of GDP is associated with a **0.03% decrease in GDP-per-capita**, holding other factors constant. This negative relationship may suggest resource dependency effects, such as the "resource curse."

c. F-test for Overall Model Significance:

Hypotheses:

- Null Hypothesis (H_0): All coefficients = 0 (no explanatory power).
- Alternative Hypothesis (H_A): At least one coefficient $\neq 0$ (model has explanatory power).

The test results (table 4) demonstrate that the F-statistic is 184.499 with a p-value of less than 0.01. Based on these results the F-test indicates that the model is statistically significant, leading to the rejection of the null hypothesis (H_0).

d. Goodness-of-Fit:

1) $R^2=0.625$:

The model explains 62.5% of the variation in log-transformed GDP-per-capita, which is substantial for cross-country economic data.

2) Adjusted $R^2=0.622$:

Adjusted R^2 is slightly lower but close to R^2 , suggesting that additional predictors do not lead to overfitting.

3) Is higher R^2 always better?

No, a higher R^2 is not always better. Overfitting can inflate R^2 without improving the model's generalizability or theoretical alignment.

4. Economic Theory:

The coefficients in Model 3 align well with economic theories. The positive and significant impact of CCE reflects the role of productivity and innovation in economic growth, supporting the endogenous growth theory. Conversely, the negative coefficient for Total-Natural-Rent/GDP underscores concerns about resource dependency, illustrating the "resource curse," where heavy reliance on natural rents hinders long-term growth. These findings are consistent with theoretical expectations and prior research, reinforcing the choice of Model 3 as the preferred model.

Part4: Further Estimation

1. Explanation for Model 4:

- **Justification for Interaction Term:** The interaction term $gee \times developing$ is included to explore how governance effectiveness (GEE) impacts GDP performance differently in developing countries compared to developed ones. Literature suggests that governance reforms may have a stronger effect in developing countries due to their greater institutional inefficiencies.
- **Health Spending (% of GDP):** Added to capture the economic impact of social investments. Research links health spending to improved workforce productivity and economic growth, aligning with SDG goals.

Estimated Equation Format for Model 4:

$$\log(\text{GDP-per-capita}) = 10.347 - 0.382 \times \text{CCE} - 0.011 \times \text{Natural-Resource-Rents} + 0.889 \times \text{GEE} - 1.324 \times \text{Developing} + 0.663 \times (\text{GEE} \times \text{Developing}) - 0.064 \times \text{Health Spending}$$

Standard Regression Results for Model 4:

Variable	Estimate	Std. Error	t-value	p-value	Significance
Intercept	10.347	0.370	27.945	<2e-16	***
CCE	-0.382	0.111	-3.432	0.0007	***
Total-Natural-Rents	-0.011	0.006	-1.718	0.0872	.
GEE	0.889	0.259	3.437	0.0007	***
Developing	-1.324	0.332	-3.987	9.13e-05	***
Health Spending	-0.064	0.024	-2.705	0.0073	**
GEE × Developing	0.663	0.219	3.022	0.0028	**

Table6: SSR for Model 4

2. Hypothesis Test for Interaction Term

- Hypotheses:
 - Null (H_0): The interaction term $gee \times developing$ has no effect ($\beta_{interaction}=0$)
 - Alternative (H_1): The interaction term $gee \times developing$ has a significant effect ($\beta_{interaction} \neq 0$)

- Results: The interaction term's coefficient ($\beta_{\text{interaction}}=0.663$) is statistically significant ($p=0.002$, significant at the 1% level).
- Interpretation: In developing countries, the impact of GEE on GDP-per-capita is amplified by 0.663 units compared to developed countries.

3. Preferred Model and Model 4:

Statistic	Model 3	Model 4
Dependent Variable	log (GDP-per-capita)	log (GDP-per-capita)
Intercept	9.214 (0.073, ***)	10.347 (0.370, ***)
CCE	1.019 (0.059, ***)	-0.382 (0.111, ***)
Total Natural Rents	-0.030 (0.009, ***)	-0.011 (0.006, .)
GEE	-	0.889 (0.259, ***)
Developing	-	-1.324 (0.332, ***)
Health Spending	-	-0.064 (0.024, **)
GEE * Developing	-	0.663 (0.219, **)
Sample Size	224	224
R2	0.625	0.8661
Adjusted R2	0.622	0.8624
Residual Std. Error	0.735 (df=221)	0.4436
F-Statistic	184.499 (df=2, ***)	233.856 (df=6, ***)

Table7: Model 3 and model 4 comparison

- Goodness-of-Fit: Model 4 achieves higher R^2 (86.61%) and adjusted R^2 (86.24%) compared to Model 3, indicating a better fit and reduced omitted variable bias.
- Significance of Variables: Model 4 provides additional insights through the interaction term and health spending, revealing nuanced relationships.
- Improved Interpretation: The interaction term in Model 4 highlights how governance effectiveness impacts GDP more strongly in developing countries, offering valuable policy implications.

4. Final Model:

Model 4 is preferred due to its higher explanatory power, as shown by increased R^2 and adjusted R^2 , which reduce bias from omitted variables. It captures variations between developing and

developed countries through the interaction term and incorporates health spending, aligning with SDGs. Overall, Model 4 not only provides a better statistical fit but also offers richer and more reliable insights for policymaking.

Part5: Conclusion

The analysis reveals that governance quality and effectiveness are key drivers of GDP performance, with a notable impact in developing countries. Natural resource dependency negatively affects growth, while NFDI and governance reforms have positive impacts. Health spending, though slightly negative in this model, reflects a trade-off between short-term costs and long-term benefits.

Considering these findings, developing countries are advised to improve governance effectiveness and invest in robust health systems to enhance workforce productivity and overall economic stability. However, this study has limitations, including possible omitted variable bias, measurement errors in governance indicators, and undiscovered collinear relationships. Future research could address these by using time-series data to explore dynamics and expanding datasets to include more governance and social variables.

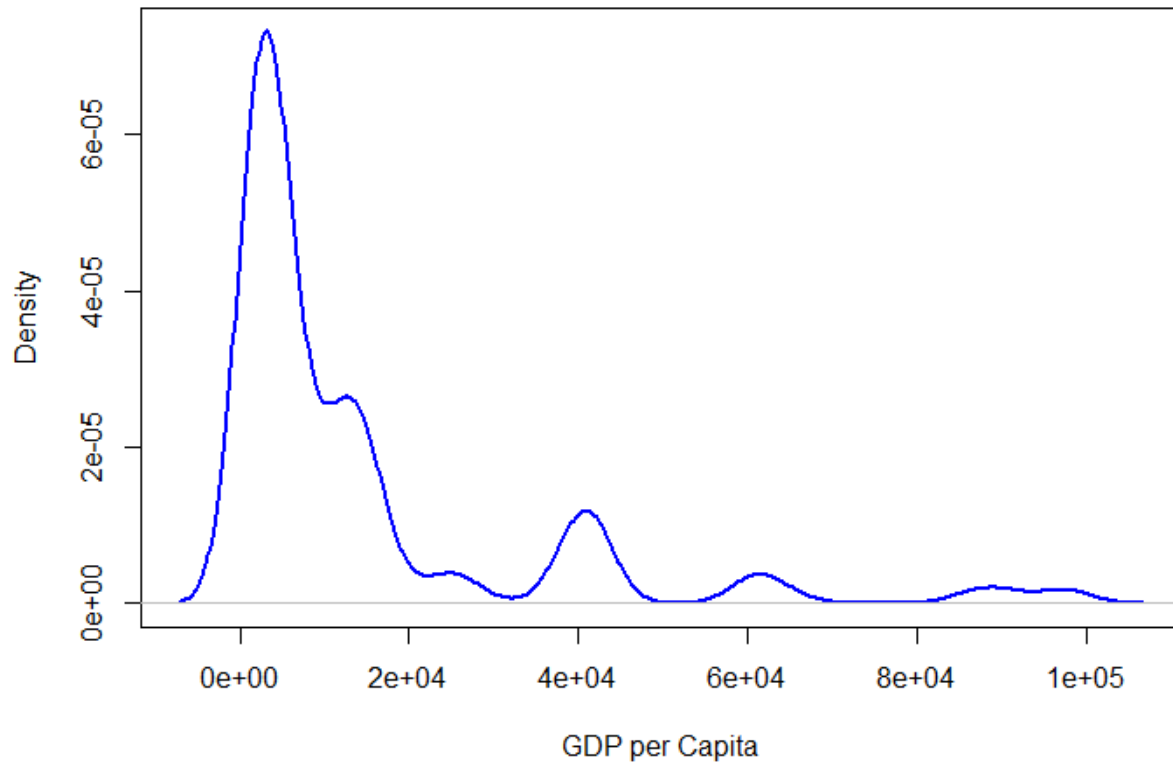
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APPENDIX:

Appendix 1: GDP-per-capita Before Log Transformation

Density Plot of GDP per Capita



Appendix 2: GDP-per-capita After Log Transformation

